higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

DATA COMMUNICATION NETWORKING
NQF LEVEL 4

(10041014)

18 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 11 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 How long is an IPv6 address?

A 32 bits
B 128 bytes
C 64 bits
D 128 bits

1.1.2 What protocol is used to send a destination network unknown message back to the originating hosts?

A TCP
B ARP
C ICMP
D BootP

1.1.3 Data communication is the transfer of messages through …

A a transmission medium.
B a linear medium.
C a road network.
D protocols.

1.1.4 What protocol does PPP use to identify the network layer protocol?

A NCP
B ISDN
C HDLC
D LCP

1.1.5 Which protocol does DHCP use at the transport layer?

A IP
B TCP
C UDP
D ARP

1.1.6 A stub network is a network …

A with more than one exit point.
B with more than one exit and entry point.
C with only one entry and no exit point.
D that has only one entry and exit point.

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1.1.7 Which ONE of the following is NOT a WAN protocol?

A Frame relay
B PPP
C Ethernet
D ATM

1.1.8 If the central hub goes down in a star topology, it affects …

A one node.
B no node.
C the entire network.
D only half of the network.

1.1.9 Which THREE statements are true about the operation of a full-duplex Ethernet network?

1. There are no collisions in full-duplex mode.
2. A dedicated switch port is required for each full-duplex mode.
3. Ethernet hub ports are preconfigured for full-duplex mode.
4. In a full-duplex environment, the host network card must check for the availability of the network media before transmitting.
5. The host network card and the switch port must be capable of operating in full-duplex mode.

A 1, 2, and 5
B 2 and 4
C 2, 3 and 4
D 5 only

1.1.10 What type of RJ45 UTP cable is used to connect a PC to a PC?

A Straight-through cable
B Crossover cable
C Console cable
D Rollover cable

(10 × 1) (10)

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1.2 Complete the following sentences by writing only the missing word or words next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 A/An … system combines many different parts in a way that works well.

1.2.2 … is a standard that uses PSTN to bring internet services mostly to homes but can also be used by businesses.

1.2.3 In … data transmission, bits are transmitted one at a time on a single channel.

1.2.4 In … more than one light signal can pass through the cable at any given time.

1.2.5 … are those activities that happen automatically on the network at a particular time.

1.2.6 … is a document that contains all the requirements that a user can expect of a product or service.

1.2.7 An application that can use network resources is known as a … application.

1.2.8 The term software package refers to an entire … of applications, such as Microsoft Office.

1.2.9 To … a network involves making small changes to the way in which the network runs and checking those small changes against a most recent baseline.

1.2.10 … transmits data across a point-to-point connection byte by byte.

(10 × 1) (10)

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1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

1.3.1 In a star topology, a computer is used as a central device.

1.3.2 Encryption coverts plain text messages into a code that needs to be decrypted in order to be read.

1.3.3 An IPv6 address is four times larger than an IPv4 address.

1.3.4 Windows server 2016 is NOT a network operating system.

1.3.5 The OSI model consists of FIVE layers.

1.3.6 Full-duplex transmissions take place in both directions at the same time.

1.3.7 The internet is defined as a network of networks.

1.3.8 The five nines uptime means a network uptime of 99.999%.

1.3.9 Cyber security is NOT important for any company's computer network.

1.3.10 A MAC address is also known as a physical address.

(10 × 1) (10)

1.4 Choose an item from COLUMN B that matches the description in COLUMN A. Write only the letter (A–M) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.4.1	Process of looking for problems, diagnosing them and correcting them	A	routing protocol
1.4.2	A document containing the security policy that is implemented in the network	B	VOIP
1.4.3	A protocol that finds the best path to deliver data	C	buffer
1.4.4	An account with unlimited rights to a network	D	diagnostic software
1.4.5	Temporary storage area that a device uses to hold incoming data before it can be processed	E	feasibility study
1.4.6	The process that involves finding out if it is possible to meet the user requirements taking into consideration time and money	F	troubleshooting
1.4.7	Specialised programmes that can probe and monitor a system	G	data link
1.4.8	The reduction in amplitude and intensity of a signal.	H	finding protocol
1.4.9	A transmission technique used to make phone calls over a data link	I	denial-of-service
1.4.10	A type of attack that prevents the legitimate use of a network or device	J	attenuation
		K	routing table
		L	administrator
		M	security policy

(10 × 1) (10)

[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

Study the pictures below and answer the questions.

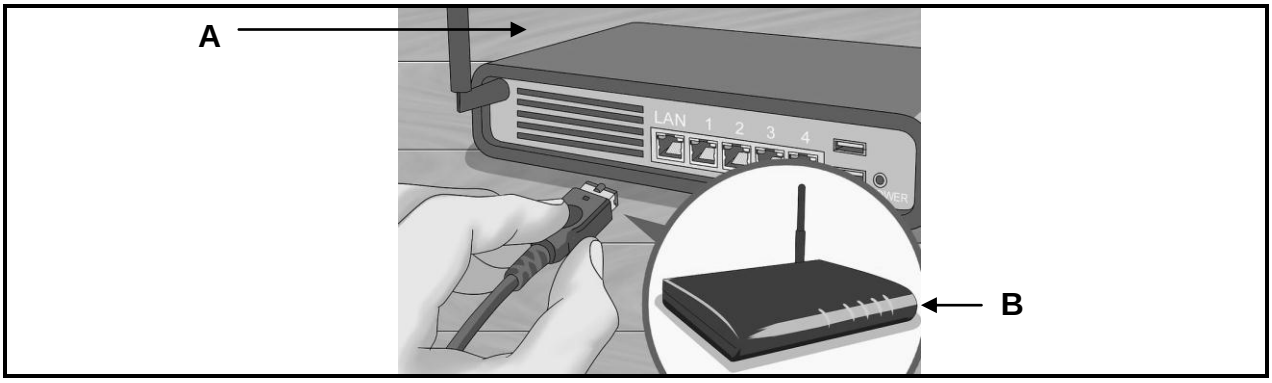


FIGURE 1

2.1 Name the devices marked A and B in FIGURE 1. (2)

2.2 Give TWO functions for each of the devices depicted in FIGURE 1 above. (4)

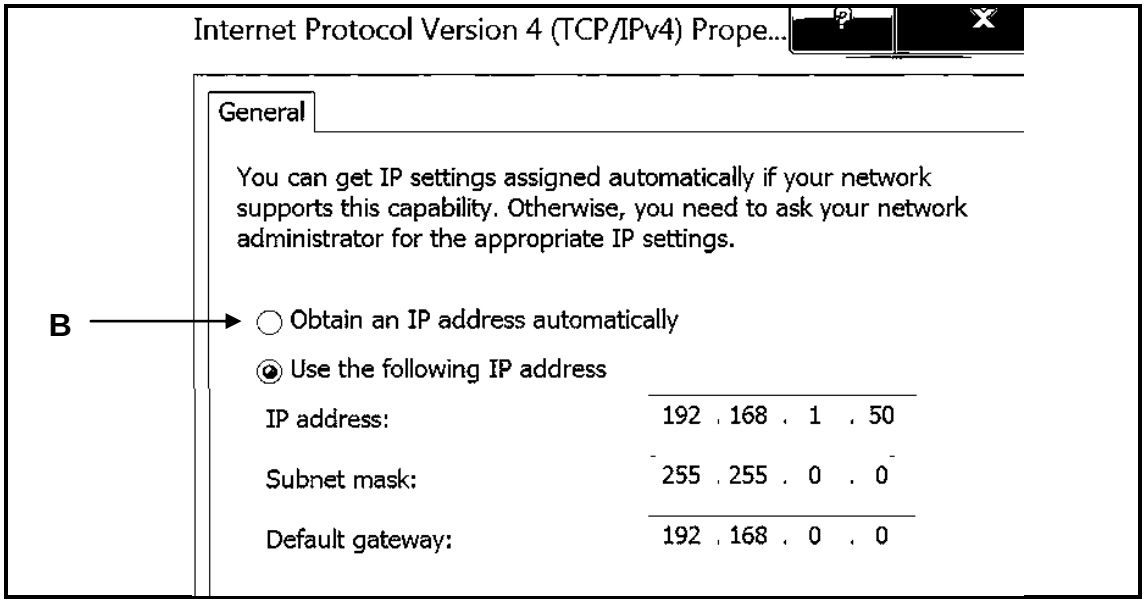


FIGURE 2

2.3 Explain in which situation will an administrator use option B. (2)

2.4 Name the host configuration protocol which is normally used with option B. (1)

2.5 Using the information given in FIGURE 2, write down the next valid IP address for another PC connected to the same network. (2)

2.6

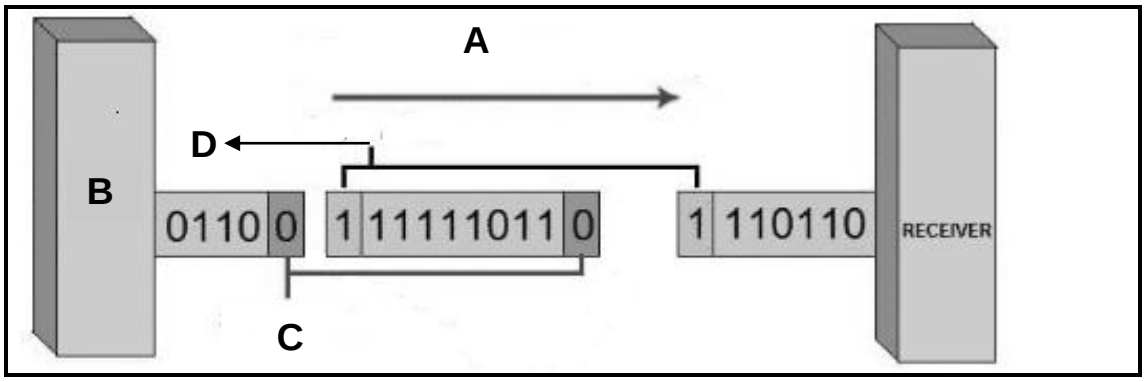


FIGURE 3

Complete the diagram shown above in FIGURE 3 by naming the letters indicated. Write only the answer next to the letter (A–D) in the ANSWER BOOK. (4)

2.7 Make a sketch to show the communication that allows data signals to be sent and received with specific timing. (4)

[19]

QUESTION 3

3.1 All data link layer protocols perform FOUR basic functions in communication networks. Name and explain each of the FOUR functions of data link layer protocols. (4 × 2) (8)

3.2 Briefly discuss the following terms that are currently being used within the IT industry globally:

3.2.1 Internet of Things (IoT)

3.2.2 Cloud computing

3.2.3 Smart personal assistants

3.2.4 Wearable technology

(4 × 2) (8)

[16]

QUESTION 4

4.1 List SIX situations or disasters that can make a network fail. (6)

4.2

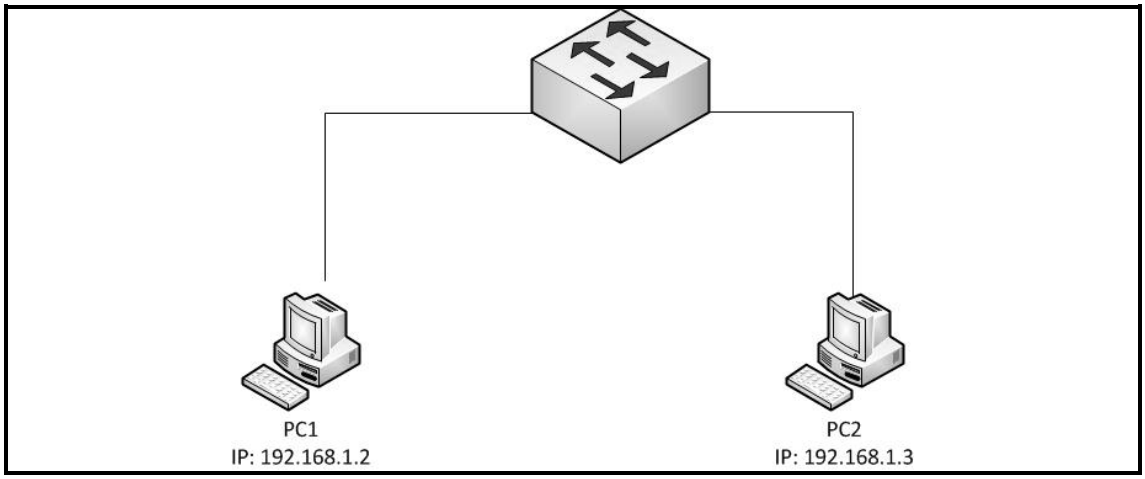


FIGURE 4

In the FIGURE 4 above, two PC's are connected to a switch. The IP addresses of the PC's are displayed in the diagram. PC1 (192.168.1.2) is connected to port 2 on the switch and PC2 (192.168.1.3) is connected to port 3 on the switch.

4.2.1 Does the switch need an IP address for PC1 to communicate with PC2? State 'Yes' or 'No' and motivate your answer. (2)

4.2.2 Write down the name of the table the switch will build (1)

4.3 You are a systems administrator for a company that has 200 employees. You need to take an important service offline temporarily, for scheduled maintenance and upgrade.

Briefly explain how you will schedule the installation so as to minimise disruptions to users. (9)
[18]

QUESTION 5

5.1 Identify FIVE tasks that are commonly performed by login scripts. (5)

5.2 Briefly describe the process of manually or statically assigning an IP address on a PC running any version of Windows operating system. (4)

5.3 List the steps involved in shutting down a Windows computer remotely using CMD (command prompt). (8)
[17]

QUESTION 6

You have been approached by a company that has expanded its offices into different countries around the world. The company wants to connect these offices without investing large amounts of money purchasing equipment. The company also wants to connect these offices in the shortest possible time. The CEO has read about virtual private networks (VPNs), remote desktops and cloud computing technologies which allow for instant information and file sharing at low cost.

The CEO has asked you to critically evaluate these three technologies, namely VPNs, remote desktops and cloud computing so that the company can make an informed decision on how best to connect its offices in different countries. **[10]**

QUESTION 7

7.1 List FIVE elements of data communication. (5)

7.2 Explain the difference between a local area network and a wide area network. (4)

7.3 The data-link layer has TWO sublayers.

7.3.1 Identify these TWO sub layers. (2)

7.3.2 Give a brief explanation of each of the layers identified in 7.3.1 (6)

7.4 Define the term broadband. (3)
[20]

QUESTION 8

8.1 Specify EIGHT steps which are carried out when installing a LAN. (8)

8.2 Name TWO advantages of fibre optic cabling over copper cabling. (2)
[10]

TOTAL SECTION B: 110
GRAND TOTAL: 150

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